Back pain? 10 possible causes and what to do about it

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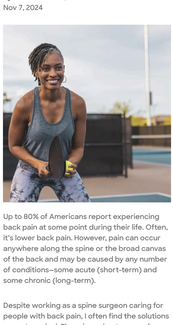

Up to 80% of Americans report experiencing back pain at some point during their life. Often, it's lower back pain. However, pain can occur anywhere along the spine or the broad canvas of the back and may be caused by any number of conditions—some acute (short-term) and some chronic (long-term).

Despite working as a spine surgeon caring for people with back pain, I often find the solutions are not surgical. There is a robust range of treatments that can effectively address many of the most common causes of back pain, before considering surgery.

Here's a look at some of the most common causes of back pain and what to know about each.

10 common causes of back pain

1. Muscle or myofascial strain

This is a very frequent source of lower back pain. It can be caused by anything from sleeping incorrectly to lifting something improperly. Strains occur when one of the muscles running on either side of the spine is injured. The muscles of the spine are used throughout the day in virtually all motions. Therefore, these muscles don't get a lot of rest, and these injuries can remain painful and linger.

2. Annular tear

An annular tear happens when the outer layer of the spinal disc, called the annulus, develops a tear, leading to significant pain.

3. Trochanteric bursitis

Trochanteric bursitis is a hip issue yet can be confused with a spinal condition. There is a tiny sac of fluid on the outside points of the hip, as well as a muscle running from the lower back around the hip points and down the thigh. When the sac of fluid becomes inflamed, it can cause pain around the hip, which sometimes radiates up towards the back.

4. Disc herniation

Disc herniation occurs when a piece of disc, which acts as a cushion between the bones of the spine, slips out of place and hits a nerve. In addition to back pain, a herniation may cause a shooting pain that goes down the buttocks and thigh. It may take two or three months, but about 90% of disc herniations repair themselves without surgery.

5. Lumbar disc degeneration (spinal stenosis)

Stenosis happens when the roots of nerves that run down the spinal column get pinched between a disc that has been worn down and an area of arthritis in the back. As with a herniation, it is often accompanied by shooting pain in the buttocks, thigh or further down the leg. Pain often subsides when sitting or laying down.

6. Arthritis

Arthritis is a degenerative condition that affects the joints in your back and can be a very common cause of back pain. Due to aging and wear and tear, the joints on the back of the spine can lose cartilage, causing bone-on-bone arthritis.

7. Scoliosis

The anatomical purpose of the spine is to keep the head in line with the hips. Scoliosis—or curvature of the spine (either front, back or sideways)—can result in the muscles of the back trying to compensate to keep the head aligned. Once the muscles become fatigued, back pain follows.

8. Osteoporotic fractures

The elderly and those with low bone density are at greater risk of bones in the spine becoming weak and brittle, a condition known as osteoporosis. At that point, any minor fall, impact or daily use can cause the bones in the spine to experience a compression fracture.

9. Spondylolisthesis

The spine is composed of blocks of bone stacked on top of one another. When one block slips forward from the one above it, it can result in back pain. There are several types of spondylolisthesis, but it usually only takes a small shift forward (a few millimeters) to pinch a nerve and cause not only pain in the back, but in your buttocks or legs as well.

10. Nerve pinching

A pinched nerve in your back can cause significant pain. This occurs when tissue or bone in the spine compress (or "pinch") nearby nerves. Often, the pain will go away on its own, but sometimes additional treatment is required.

3 lifestyle factors that can cause lower back pain

Lifestyle factors play a significant role in the development of back pain, and can make any existing back pain problems much worse.

1. Weight gain

The spine is a stack of bones supported by muscles, so when there's excess weight on the body, those muscles have to work harder, leading to strain, injury or earlier disc degeneration.

2. Inactivity

In the modern world, prolonged periods of sitting can contribute to back pain as well. Inactivity weakens core muscles, which are crucial for supporting the spine. Strengthening these muscles through exercises like Pilates and yoga can prevent or alleviate back pain.

3. Pregnancy

Many pregnant women also experience lower back pain, which is caused by weight gain, as well as changes to your pelvis and center of gravity. Pregnancy back pain can often be improved by stretches, exercise and rest.

Treatment options for lower back pain

Although back pain is rarely life-threatening, anyone who has experienced it knows how easily it can spoil day-to-day life. Though surgery may be the answer in some cases, the first step is often diagnosing the cause of the pain through non-invasive methods.

Nonsurgical treatments for lower back pain

There are many nonsurgical treatments available for managing lower back pain:

Physical therapy: Strengthening the muscles around the spine, especially the core, can significantly reduce pain.

Injections and ablations: Targeted injections into the facet joints or around the spinal discs can help diagnose and relieve pain.

Lifestyle changes: Losing weight, improving your posture and engaging in regular exercise can prevent lower back pain from starting, or help it improve.

Lower back pain can be a complex issue, but many treatment options are available. Seeking medical advice early on, especially when pain begins to interfere with daily life, can lead to better outcomes. With proper diagnosis, lifestyle changes and non-invasive treatments, many people can find relief without resorting to surgery.

About the Author

Dr. James Rizkalla completed his orthopedic surgery residency at Baylor University Medical Center in Dallas, Texas, followed by two fellowships in complex spinal surgery. He is currently the Medical Director for orthopedic research at Baylor University Medical Center.
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